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Mobile Survey Design Research
Experiment 16: Date Entry Method

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Abstract

The present study compared usability among three date entry designs: Picklist, keyed-entry, and hybrid. The findings show that the hybrid design has the lowest date entry accuracy rate, though accuracy is generally high across all three designs. Date entry completion time is longer for the picklist design compared to both keyed entry and hybrid. Date entry is generally easy with any of the three designs though the keyed-entry design appears least easy; and the picklist and keyed-entry designs are almost equally preferred by most participants, with very few preferring the hybrid design.

Keywords: Mobile Survey, Date entry, Picklist, Keyed-entry, Hybrid

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Lin Wang: study supervision, analysis and interpretation of results, manuscript preparation

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Introduction

1. Purpose

To investigate the optimal date entry method among those commonly used in mobile Web surveys.

Specifically, three designs of date entry were compared: picklist, keyed-entry, or hybrid of picklist and keyed-entry. The picklist design refers to a design in which a date component is “picked” from a list of options, for example, a month entry was picked from a list of 12 months. The keyed-entry design refers to a design in which a date component is typed into a text field as usual text typing. The hybrid design refers to a combination of picklist and keyed-entry in which a date component is “picked” from a list of options while another date component was typed in a text field, for example, a month entry was picked from a list of 12 months while a year entry was typed in.

2. Specific Research Question

Do respondents enter dates more accurately, faster, and with better satisfaction, using a picklist, a keyed-entry, or a hybrid format?

Methods

1. Participants

Thirty participants participated in the study and their demographics are summarized in Table 1. Seventy-three percent of participants reported using a smartphone for more than two years, 10% between one and two years, 17% between 6 months and one year. Seventy percent of participants reported using a smartphone every day, while the other 30% used their smartphone less than daily. The participants reported their familiarity with smartphone being an average of 3.6 ($SD = 0.8$) on a scale of 1 to 5, where higher scores represented more familiarity. The participants had habitual near vision of 20/50 or better (sufficient for reading newspaper) and had no color vision impairment.

All participants received a Privacy Act Statement that states the purpose of the study, information to be collected, and the authority under which the information is collected. A written consent to data collection was obtained from all participants. The data collection was approved by the U.S. Office of Management and Budget. Each participant received a monetary incentive for participating in the study. All participants were from the metropolitan area of Washington DC and were recruited at community centers.

Table 1. Participants Demographics

Sex (number of participants)	
Male	4
Female	26
Age in years	
Mean (SD)	70 (3.6)
Range	62 - 77
Education (number of participants)	
Completed high school	5

Some college, no degree	7
Associate degree	2
Bachelor degree	9
Post graduate degree	7
Hispanic origin (number of participants)	
Yes	1
No	29
Race (number of participants)	
White	17
Black or African American	8
Asian	5

2. Screen Design

Three versions of the mobile survey were designed using different date-entry formats: (1) picklist only (Figure 1, A), (2) keyed-entry only (Figure 1, B), and (3) a hybrid of picklist and keyed-entry (Figure 1, C).

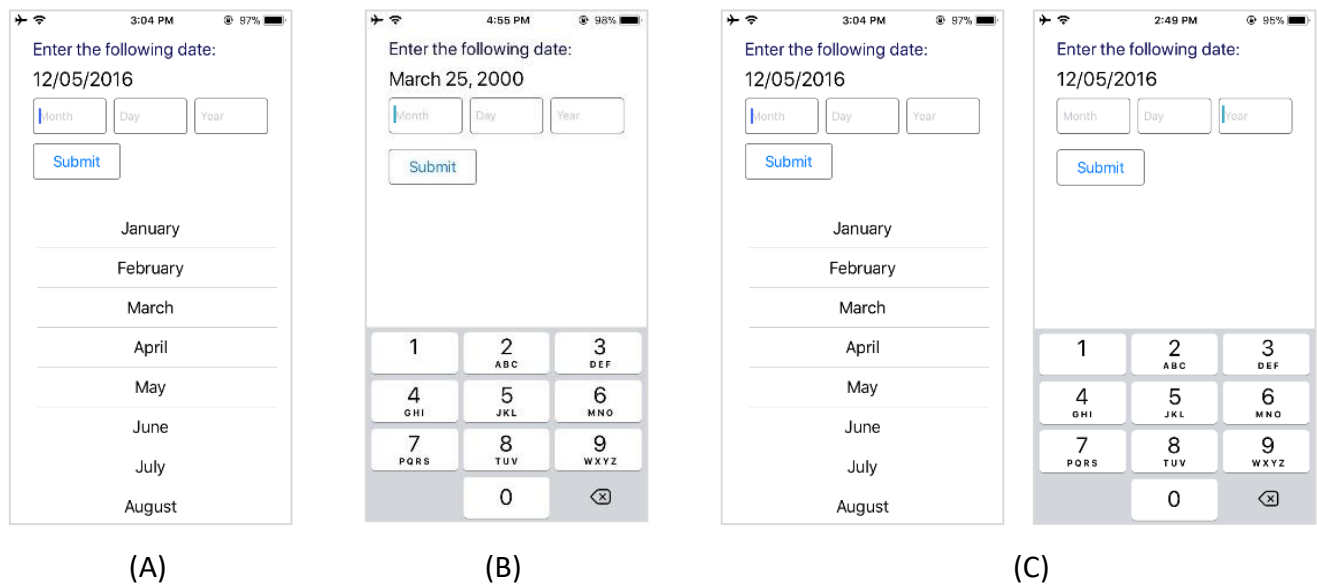


Figure 1. Panel A: an example of picklist design. The month entry is to be entered by picking a month from the picklist. Panel B: an example of keyed-entry design. The month entry is to be entered by typing text in the text fields. Panel C: an example of hybrid design. The month entry is to be entered by picking a month from the picklist, while the day and year entry is typed in the text field.

3. Performance Measures

- Completion Time:** Length of time it takes the participant to enter data, from the time any text field or picklist is activated until the time the “next” button is selected.
- Accuracy:** A binary variable indicating whether final date entered was correct or not.

- c. *Difficulty rating*: Self-report of task difficulty on a 5-point scale with 1 being very easy and 5 being very difficult.
- d. *Preference*: Preferred date entry format among the six designs.

4. Experimental Factors

a. Date-Entry Type

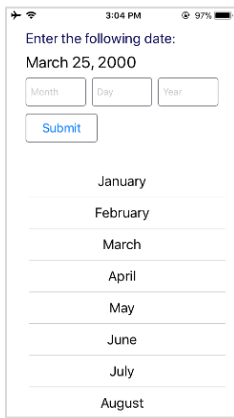
Level 1: Picklist

Level 2: Keyed-entry

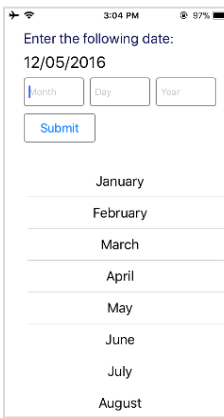
Level 3: Hybrid – The combination of Picklist and Keyed-entry (month: picklist, day: keyed entry, year: keyed-entry).

5. Experimental Design

This was a within-subjects design with one experimental factor, *Date-Entry Type* (Picklist, Keyed-entry, Hybrid). Under each design, the participant was given dates in a congruent format (Figure 2, A) with date entry design in half of trials and incongruent in the other half of trials (Figure 2, B). The order of presentation of the 3 designs and congruency was counterbalanced across participants.



(A)



(B)

Figure 2. A: an example of congruent design for hybrid condition. The month is presented as a word and is to be entered by picking a month from the picklist. B: an example of incongruent design for hybrid condition. The month is presented numerically and is to be entered by picking a month from the picklist.

6. Experimental Procedure

- a. The participant was presented with a date on the screen and read it aloud to ensure accurate perception.
- b. The participant entered the date directly beneath the one presented on the screen using picklist, keyed-entry, or hybrid.
- c. The participant tapped the 'next' button which brought up a new screen with another date.
- d. After the survey was complete, the participant rated the ease of completing the task on a 5-point Likert scale where '1' was *very easy* and '5' was *very difficult*.
- e. The participants were shown all six date-entry designs and asked to select a preferred design.

7. Data Analysis

Completion Time: We conducted a repeated measures ANOVA with Date-Entry Type (Picklist, Keyed-entry, Hybrid) as the within-subjects factor and Completion Time as the outcome.

Accuracy: We used generalized estimating equations (GEE) to fit a repeated measures logistic regression, predicting accuracy from date-entry type.

Difficulty rating: We used generalized estimating equations (GEE) to fit a repeated measures logistic regression, predicting difficulty from date-entry type.

Preference: We analyzed the frequencies of participants' choice to determine which design was most preferred.

Results

Completion time: The results of the repeated measures ANOVA showed an effect of Date-Entry Type on completion time ($F(2, 27) = 8.24, p < .002$). The results indicated longer time on task in the picklist format ($M = 28.25$ sec, $SD = 13.97$) compared to both the keyed-entry format ($M = 22.82$ sec, $SD = 10.97, p = .024$) and hybrid format ($M = 20.55$ sec, $SD = 7.63, p = .0004$). The difference in time on task between the keyed entry and hybrid conditions was not statistically significant ($p = .15$).

Accuracy: Date entry correct rate by condition is shown in Table 2. While there was a high rate of correct date entry across conditions ($> 92\%$), we used generalized estimating equations (GEE) to fit a repeated measures logistic regression, predicting accuracy from date-entry type. Compared to the hybrid design, the odds ratio of an accurate response was significantly higher when the date entry format was picklist or keyed-entry, $b = 1.00$ (.39), 95% CI [.23, 1.77], $p = .01$ and $b = 1.57$ (.46), 95% CI [.67, 2.47], $p = .0007$, respectively.

Table 2. Date entry correct rate by condition

Condition	Correct Rate (%)
Picklist	97.1
Keyed-entry	98.3
Hybrid	92.5

Difficulty rating: Participants rated the task as easy overall, with 70% giving either a 1 or 2 (lowest difficulty levels). See Table 3. We used generalized estimating equations (GEE) to fit a repeated measures logistic regression, predicting difficulty from date-entry type. Compared to hybrid design, the odds of a higher difficulty rating were greater when the date entry format was keyed entry, $b = .33$ (.12), 95% CI [.09, .57], $p = .007$.

Table 3. Distribution of difficulty ratings by condition

Date-entry type	Difficulty Scale				
	1	2	3	4	5
Picklist (n, %)	13, 43.3%	8, 26.7%	8, 26.7%	0, 0%	1, 3.3%
Keyed entry (n, %)	11, 36.7%	8, 26.7%	10, 33.3%	0, 0%	1, 3.3%
Hybrid (n, %)	16, 53.3%	7, 23.3%	6, 20.0%	1, 3.3%	0, 0%

Preference: Forty three percent of participants preferred picklist, 40% preferred keyed-entry, 7% preferred hybrid, and 10% did not make a choice.

Table 4. Summary of the outcomes

	Picklist	Keyed-entry	Hybrid
Date entry accuracy (%)	97.1	98.3	92.5
Date entry time (sec)	28.25	22.82	20.55
Easiness (% saying 1 or 2)	70	63.4	76.6
Preference	43	40	7

Recommendation

The present study compared usability among three date entry designs: Picklist, keyed-entry, and hybrid. The findings show that the hybrid design has the lowest date entry accuracy rate, though accuracy is generally high across all three designs. Date entry completion time is longer for the picklist design compared to both keyed entry and hybrid. Date entry is generally easy with any of the three designs though the keyed-entry design appears least easy; and the picklist and keyed-entry designs are almost equally preferred by most participants, with a few preferring the hybrid design.

We recommend keyed-entry format as the first choice for date entry given its effectiveness and general efficiency over picklist. Hybrid is still a viable option especially when efficiency is of import.